Publish the Rest: Research Funded by Trading Rents in a Kyle-Back Market

Keep private only the Gaussian part that pays the research bill

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Abstract

Research paid for out of an informed trader's profits raises the question of how much of its output should stay private, and in what form. In a Kyle-Back market where liquidity traders decide whether to participate, the answer depends on whether part of the output can be published. If all of it must stay private, a coarse threshold signal can maximize welfare over every experiment with a given information budget, and in a worked example every Gaussian private signal is worse than doing no research. If the producer can publish, the optimum over all joint public and private experiments keeps private only a Gaussian component whose trading rent exactly covers the research bill and publishes the rest. This split loses the least liquidity, makes prices as informative as possible at every trading date, and turns extra information capacity into public accuracy. When research costs a fixed amount plus a price per nat, its optimal quantity solves a strictly concave problem in which each unit of cost is scaled up by the liquidity lost to finance it.

1 Introduction

Much research is financed by the trading profits it makes possible. A data vendor with an exclusive client, an analyst whose work pays for itself through a fund's positions, or a research desk inside a trading firm all face the same choice. They can keep their output private, which earns a trading rent but drives liquidity traders away, or publish it, which improves prices but removes the rent that pays for the work. This note solves that choice in a small Kyle-Back model with endogenous participation of liquidity traders, and the answer turns on the shape of the information as well as its amount.

With all output private, the best signal is coarse. A previous note in this series (Equilibrium Daily No. 0017) shows that information explaining a share r of fundamental variance earns the insider between r and $\sqrt{r}$ of the full-information profit, with ordered categories at the bottom and normal noise at the top. When the insider's profit is a loss to liquidity traders who can stay away, the low end is attractive. A threshold signal delivers the forecasting value r with the smallest possible adverse selection, and it can be the global optimum over all experiments (Proposition 1). In the worked example below every Gaussian private signal does worse than no research at all.

With publication allowed, the ranking reverses. Public information improves prices without creating adverse selection, so the private component's only remaining job is to earn enough rent to cover the research bill. Rent per unit of private variance is largest for normal information, so the optimal private component is Gaussian and as small as the bill allows, and everything else is published (Theorem 1). The optimum is global over all joint experiments, including randomized and state-dependent disclosure, and the same split also makes prices as informative as possible at every interim trading date. The coarse signal that is optimal when information must stay private is therefore the wrong guide to disclosure policy.

The model is deliberately small. It assumes a committed and observable research contract, one licensee, liquidity traders with uniformly distributed benefits, risk-neutral competitive market makers, and selection of the high-participation equilibrium. Section 5 lists these assumptions and the closest literature.

2 Model

The fundamental $V \sim N(0, \sigma_v^2)$ is paid at date 1, and in Sections 3 and 4 we normalize $\sigma_v = 1$. A single risk-neutral trader receives a signal with posterior mean $m = \mathbb{E}[V \mid S]$, and competitive risk-neutral market makers set prices equal to conditional expectations given total order flow, as in Back (1992). Write $r = \mathrm{Var}(m)/\sigma_v^2$ for the share of fundamental variance the signal explains.

Lemma 1 (Equilibrium No. 0017)**.** *With noise order flow of volatility σ_u, the insider's expected profit is $\sigma_u \sigma_v a$, where $a = \max \mathbb{E}[(m/\sigma_v)\,\xi]$ over couplings of m with a standard normal ξ, and $r \le a \le \sqrt{r}$. A monotone coarse signal (ordered categories or a threshold) attains $a = r$, and a normal signal attains $a = \sqrt{r}$. The law of m is feasible for some signal if and only if m/σ_v lies below the standard normal in convex order.*

Participation. A unit mass of liquidity programs has gross benefits uniformly distributed on $[0, B]$. A participating mass n generates noise order flow with variance rate $\sigma_0^2 n$, so $\sigma_u = \sigma_0\sqrt{n}$. Write $\kappa = \sigma_0\sigma_v$. The insider's profit $\kappa a\sqrt{n}$ is a loss shared by participants, $\kappa a/\sqrt{n}$ each, and the marginal participant has benefit $B(1-n)$. Entry therefore requires

$$\kappa a = B\sqrt{n}\,(1-n). \tag{1}$$

The right side rises up to $n = 1/3$ and then falls, so there are two positive equilibria when $a < 2B/(3\sqrt{3}\,\kappa)$ and none above that threshold. We select the high equilibrium, which is stable under entry best responses. At the high equilibrium the insider's rent is $\Pi = Bn(1-n)$ and liquidity programs' gross benefits are $G(n) = B(n - n^2/2)$. Rents are transfers, and G is the real gain from trade.

Equal accuracy, different markets. At fixed r a coarse signal minimizes a and a normal signal maximizes it, so on the open region

$$\tfrac{3\sqrt{3}}{2}\, r < B/\kappa < \tfrac{3\sqrt{3}}{2}\,\sqrt{r}$$

the market operates with coarse information and has no positive-participation equilibrium with equally accurate normal information. At $r = 2/\pi$, $\kappa = 1$ and $B = 1.9$, the sign of the fundamental supports participation $n = 0.547$, while the equally accurate normal signal has $a = 0.798$, above the threshold 0.731.

Research and welfare. A neutral producer commits to an experiment and charges a fixed real cost F to one licensee, who buys if the anticipated rent covers F. A real decision maker observes the terminal price and chooses x to maximize $\mathbb{E}[xV - x^2/(2\chi)]$, so price information is worth ηr with $\eta = \chi\sigma_v^2/2$. Welfare conditional on production is

$$W = G(n) + \eta r - F,$$

and without research $n = 1$ and $W = B/2$. Information is limited by a budget $I(V; \text{output}) \le k$ nats.

3 When all research must stay private

Proposition 1. *If the whole output goes to the licensee, every experiment satisfies*

$$W \le B(n - n^2/2) + \frac{\eta B}{\kappa}\sqrt{n}\,(1-n) - F.$$

The right side is strictly concave in n. If its maximizer n^* satisfies $Bn^*(1-n^*) \ge F$, the implied $a^* = B\sqrt{n^*}(1-n^*)/\kappa$ is at most $2/\pi$, and the threshold signal with $r = a^*$ has entropy below k, then that threshold signal attains the bound and is optimal over all experiments.

Proof. Combine $r \le a$ from Lemma 1 with (1). A threshold signal is monotone coarse, so it has $r = a$, and its r ranges over $[0, 2/\pi]$ as the threshold moves, which makes the bound attainable under the stated conditions. □

Take $B = \kappa = \eta = 1$, $F = 0.1$ and $k = \log 2$. The optimum has $n = 0.5545$ and $r = a = 0.3317$, attained by the signal $\mathbf{1}\{V > 1.312\}$, which uses 0.313 nats and earns the licensee $0.247 > F$. Welfare is 0.6325, against 0.5 without research. A Gaussian private signal has $a = \sqrt{r}$, and substituting into the entry condition gives

$$W_{\text{Gauss}} - 0.5 = (1-n)^2(n - \tfrac{1}{2}) - 0.1 \le \tfrac{1}{54} - 0.1 < 0,$$

so every Gaussian private experiment is worse than no research. The general idea that an informed party restrains information to protect liquidity is in Mendelson and Tunca (2004); the increment here is the global bound over all signal shapes at a given accuracy.

4 When the producer can also publish

The producer now commits to a joint experiment (D, S) with $I(V; D, S) \le k$. On purchase, D is published and S goes privately to the licensee; liquidity programs enter after seeing D. Define

$$n_F = \frac{1 + \sqrt{1 - 4F/B}}{2}, \qquad d_F = \frac{F^2}{\kappa^2 n_F}, \qquad R_k = 1 - e^{-2k}.$$

Theorem 1. *Research can be funded if and only if $F \le B/4$ and $d_F \le R_k$. Over all joint experiments, maximal welfare with production is*

$$W^* = B\big(n_F - n_F^2/2\big) + \eta R_k - F.$$

It is attained by independent normal components A, Q, E with variances $R_k - d_F$, d_F and $1 - R_k$ and $V = A + Q + E$, publishing A and giving Q to the licensee. If the real decision is taken at an interim date t, the same split maximizes $\mathrm{Var}(P_t)$ *and welfare becomes $G(n_F) + \eta[R_k - (1-t)d_F] - F$.*

Proof. Funding as rent. Conditional on any public realization, entry gives rent $h = Bn(1-n) \le B/4$ whatever the conditional law of V, and purchase requires $\mathbb{E}h \ge F$.

Liquidity cost of rent. For given h participation is at most $n_+(h) = (1 + \sqrt{1 - 4h/B})/2$, and the liquidity loss $L(h) = B/2 - G(n_+(h))$ is increasing and convex. By Jensen's inequality, spreading the rent burden unevenly across public states cannot help, and $\mathbb{E}G(n) \le G(n_F)$.

Private variance needed for rent. Lemma 1 applies conditionally on D, and Cauchy-Schwarz gives $h^2 \le \kappa^2 n\, \delta_D$, where δ_D is the conditional variance of the private posterior mean. Hence $\delta_D \ge d(h) = h^2/(\kappa^2 n_+(h))$, which is increasing and convex, so expected private variance is at least d_F.

Budget. Rate distortion for a normal source bounds total explained variance by R_k, so $d_F \le R_k$ is necessary and decision value is at most ηR_k. Adding the bounds gives $W \le W^*$.

Attainment. In the Gaussian split, $I(V; A, Q) = k$, the terminal price is $A + Q$, the conditional private posterior is normal with variance d_F, rent equals F and participation is n_F, so every bound binds. For the interim date, the Hermite expansion of No. 0017 applied conditionally gives $\mathrm{Var}(P_t) \le q + t\delta$, where q is public explained variance, and $q + \delta \le R_k$ with $\delta \ge d_F$ gives $\mathrm{Var}(P_t) \le R_k - (1-t)d_F$, which the split attains. □

With $B = \kappa = \eta = 1$, $F = 0.1$ and $k = \log 2$ as before, $n_F = 0.887$, the private component explains $d_F = 0.011$ of fundamental variance and the public component 0.739. Welfare is 1.144, against 0.633 at the private-only optimum and 0.5 without research. Two features matter. The private component is tiny because normal information earns the most rent per unit of variance. And as the budget k grows with F fixed, d_F and participation stay put and all additional explained variance is published.

Terminal welfare alone does not pin down the split, since a non-Gaussian public garbling of a Gaussian terminal posterior can tie it. Any positive value of information before the last trading date selects the Gaussian split, because it releases information to prices earliest.

How much research. Let research cost $C(k) = F + ck$. Since $dG/dh = -(1-n)/(2n-1)$ on the high branch, the optimal quantity maximizes the strictly concave function $G(n_+(F + ck)) + \eta(1 - e^{-2k}) - F - ck$, and an interior optimum satisfies

$$2\eta e^{-2k} = c\,\frac{n}{2n-1}, \qquad n = n_+(F + ck).$$

The marginal value of information equals its resource cost times $n/(2n-1) > 1$, the liquidity lost to finance one more unit of rent, and the multiplier grows without bound as the required rent approaches the market's capacity $B/4$. With $B = \kappa = \eta = 1$, $F = 0.02$ and $c = 0.3$, the optimum uses 0.648 nats, participation is 0.689, and of the 0.726 explained variance only 0.067 is private.

5 Scope and related work

The results describe a committed, observable research contract sold to one licensee, with no hidden substitute research, and the welfare optimum is conditional on research being financed through trading rents, since a direct transfer to the producer would remove the need for any private advantage. Participation uses uniform benefits, which gives closed forms; the proof of Theorem 1 uses only that the liquidity loss and the private-variance requirement are increasing and convex in the rent, so it extends to any benefit distribution

with those two properties. The high-participation equilibrium is selected, and a no-trade equilibrium also exists. The model says nothing about managers who disclose after observing news, about risk-bearing dealers, or about field magnitudes.

Endogenous liquidity participation and its interaction with public information are in Pagano (1989), Dow (2004) and Han, Tang and Yang (2016), and strategic restraint in information production is in Mendelson and Tunca (2004). The sale of private and public information, including noisy versus concentrated signals, is studied by Admati and Pfleiderer (1986), Garcia and Sangiorgi (2011) and Aoyagi (2025), whose analyst adjusts public precision to hold information asymmetry fixed. Viswanathan and Xing (2026) show that normal signals are optimal for a privately acquiring trader under a mutual-information cost. We have not found the global frontier over arbitrary joint experiments, its simultaneous attainment of the funding, liquidity and price-discovery bounds, or the reversal between the private-only and joint optima in the work we searched.

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